The simulation team ran several tests, and quite a few turned out to be promising, especially a three-stage vehicle with the Vikas engine as the second stage and the first and third stage with solid propulsion. This continues to be the configuration of PSLV, which has had thirty-five consecutive successful launches till 22 June 2016. The ISRO workhorse, as PSLV has come to be known, has launched more than 100 satellites, a little more than half of them for foreign countries.

When the engine was approved as part of PSLV in 1979, such a stupendous success was not even in the dreams of many scientists. I consider it a matter of luck, too, as Brahm Prakash got the Vikas engine project as the second stage of PSLV approved just before he retired. What's more, I was made the project director of the second stage. But before that, I had to pass another hurdle—and demolish yet another one.

It was still 1979, the year of SLV-3's first flight, which was a failure. The rocket was successfully launched the next year. Brahm Prakash who was on his last leg as the VSSC director, formed two committees—one to assess the level of acquisition of the Viking engine and the other to see if a semi-cryogenic engine was doable. The first panel was headed by Kalam, the second by Kurup —both hardcore solid propulsion fans. I had my reasons to believe that the 'solid' guys had influenced Brahm Prakash to put in place these committees as a last impediment to Vikas.

I voiced my concern to Brahm Prakash.

'You don't worry,' he told me. 'They wouldn't dare tell me wrong things about your engine.'

'Indeed,' I told Brahm Prakash. 'But my worry is that they wouldn't tell you the right things either.'

Brahm Prakash asked me to drop my aggression and cooperate with the committees.

Kalam called division heads and other senior scientists for periodic analysis of the feasibility of Vikas, where I, as the person who headed the project, had to be present. The initial meetings did not progress beyond the skepticism of whether we have mastered the liquid propulsion technology. And one day, I stood up.

'How are you going to find out if we have acquired the technology,' I asked the gathering of mostly solid propulsion enthusiasts. 'One way is to cross